

NORTH SYDNEY BOYS HIGH SCHOOL

Ext 1

2017 HSC ASSESSMENT TASK 3 (TRIAL HSC)

Mathematics

Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Hwang
- ☐ Mr Ireland
- ☐ Dr Jomaa
- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Ms Ziazaris

Student Number:

(To be used by the exam markers only.)

Question No	1-10	11	12	13	14	Total	Total
Mark	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{70}$	$\overline{100}$

Section I

10 Marks

Attempt Questions 1 – 10

Use the multiple choice answer sheet for Questions 1 – 10

1 $\int \frac{dx}{\sqrt{64-9x^2}}$ is equal to

- (A) $\frac{1}{9} \sin^{-1}\left(\frac{9x}{64}\right)$ (B) $\frac{1}{3} \sin^{-1}\left(\frac{3x}{8}\right)$ (C) $\frac{1}{9} \sin^{-1}\left(\frac{3x}{8}\right)$ (D) $\frac{1}{3} \sin^{-1}\left(\frac{9x}{64}\right)$

2 The domain and range of the function $f(x) = 3 \sin^{-1}(4x-1)$ are respectively

- (A) $0 \leq x \leq \frac{1}{2}$ and $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$ (B) $-\frac{1}{2} \leq x \leq 0$ and $-\frac{\pi}{2} \leq y \leq \frac{5\pi}{2}$
(C) $0 \leq x \leq \frac{1}{2}$ and $-\frac{\pi}{2} \leq y \leq \frac{5\pi}{2}$ (D) $-\frac{1}{2} \leq x \leq 0$ and $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$

3 If $y = f(x)$ is a linear function with slope $\frac{1}{2}$, then the slope of $y = f^{-1}(x)$ is

- (A) 2 (B) $\frac{1}{2}$ (C) -2 (D) $-\frac{1}{2}$

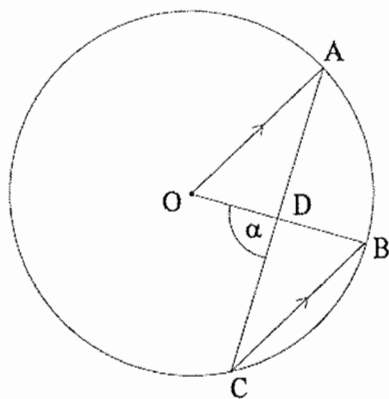
4 The derivative of $y = \cos^{-1} 3x + x \cos^{-1} 3x$ is

- (A) $\cos^{-1} 3x - \frac{x+1}{\sqrt{9-x^2}}$ (B) $\cos^{-1} 3x - \frac{3(x+1)}{\sqrt{1-9x^2}}$
(C) $\cos^{-1} 3x + \frac{x-1}{\sqrt{9-x^2}}$ (D) $\cos^{-1} 3x + \frac{3(x-1)}{\sqrt{1-9x^2}}$

5 Given that α, β, γ are roots of $3x^3 - 2x^2 + x - 1 = 0$, then $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ is equal to

- (A) 2 (B) -1 (C) 1 (D) -2

- 6 The points A , B and C lie on a circle with centre O . OA is parallel to CB . AC intersects OB at D , and $\angle ODC = \alpha$. What is the size of $\angle OAD$ in terms of α ?



- (A) $\frac{\alpha}{2}$ (B) $\frac{\alpha}{3}$ (C) $\frac{2\alpha}{3}$ (D) 3α
- 7 Find the acute angle (to the nearest degree) between the lines $2x - 3y + 6 = 0$ and $x + 2y - 12 = 0$
- (A) 7° (B) 60° (C) 14° (D) 30°
- 8 The point P divides the interval AB joining $A(-4, -3)$ and $B(1, 5)$ externally in the ratio 3:2. The coordinates of P are
- (A) $(-14, -19)$ (B) $(-11, -21)$ (C) $(11, 21)$ (D) $(14, 19)$
- 9 The equation of the normal to the parabola $x = 6t$, $y = 3t^2$ at the point where $t = -2$ is
- (A) $x - 3y + 24 = 0$ (B) $x - 2y + 36 = 0$
 (C) $2x + y + 12 = 0$ (D) $2x - y - 12 = 0$
- 10 The solution to the inequality $\frac{|3x+1|}{x} \leq 1$ is
- (A) $-\frac{1}{2} \leq x < -\frac{1}{4}$ (B) $x < 0$ (C) $x \leq -\frac{1}{2}$ (D) No solution

Section II

60 Marks

Attempt Questions 11-14

Answer each question on a NEW page. Extra writing booklets are available.

In Questions 11 – 14, your responses should include all relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW page

- (a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin 2x \cos 2x}{3x}$. 1
- (b) Find the exact value of $\cos^{-1} \cos \left(\frac{5\pi}{4} \right)$. 1
- (c) Find $\int \cos^2 9x \, dx$. 2
- (d) Solve $\sin x + \cos x = 1$ for $0 \leq x \leq 2\pi$ 3
- (e) When the polynomial $P(x)$ is divided by $1 - x^2$ it gives $4 - x$ as the remainder.
What is the remainder when $P(x)$ is divided by $1 + x$? 2
- (f) If the three roots of $x^3 - 6x^2 + 3x + k = 0$ form an arithmetic progression,
find the value of k . 3
- (g) Prove using mathematical induction that $n^3 + (n+1)^3 + (n+2)^3$ is divisible
by 9 for $n = 1, 2, 3, \dots$ 3

Question 12 (15 marks)

Start a NEW page

(a) Use the substitution $x = u^2 - 1$ for $u > 0$ to evaluate $\int_3^8 \frac{x-1}{\sqrt{x+1}} dx$. 4

(b) A bar of gold is initially at a temperature of -8 degrees Celsius.

It is taken into a nearby room where the air temperature is 22 degrees Celsius.

The rate at which the bar warms follows Newton's Law, that is, $\frac{dT}{dt} = -k(T - 22)$,

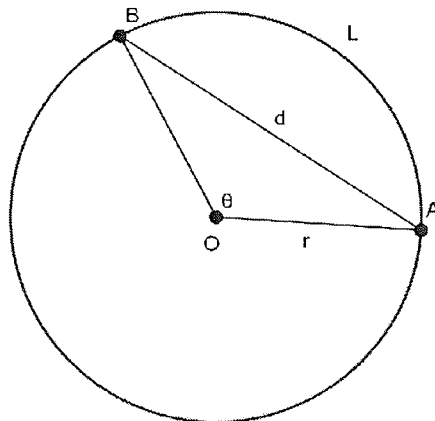
where k is a constant, time t is measured in minutes and temperature T is in degrees Celsius.

(i) Show that $T = 22 - Ae^{-kt}$ is a solution of the equation above, and evaluate A . 2

(ii) Given that the bar's temperature reaches 4 degrees Celsius in 90 minutes, find the exact value of k . 2

(iii) Find the temperature of the bar after another 90 minutes. 1

(c) An arc AB of a circle subtends an angle of θ radians at the centre of a circle of radius r . The arc's length is L , and the associated chord's length is d . (See diagram).



(i) If $L : d = 4 : 3$, show that $3\theta = 8 \sin \frac{\theta}{2}$. 3

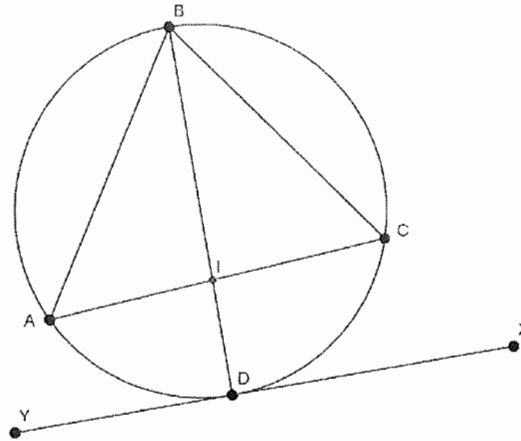
(ii) Using $\theta = 2.5$ as a first approximation, use Newton's method once to find a second approximation (to 3 decimal places) for θ . 3

Question 13 (15 marks)

Start a NEW page

- (a) In the circle, $\angle ABD + \angle BCA = 90^\circ$, and XY is tangent to the circle at D .

The chords AC and BD intersect at I . (Copy or trace the diagram in your writing booklet.)



- (i) Prove $\angle BCD = 90^\circ$ and hence that BD is a diameter of the circle. 2
- (ii) Prove that if $\triangle ABC$ is isosceles with $AB = BC$, then $AC \parallel XY$. 2
- (b) The velocity of a particle moving in a straight line is given by $\frac{dx}{dt} = 1 + x^2$, where x is the displacement in metres from the origin, and t is the time in seconds. Initially the particle is at $x = 1$.
- (i) Find an expression for the acceleration of the particle in terms of x . 2
- (ii) Find an expression for the displacement of the particle in terms of t . 2
- (c) Assume that a spherical snowball melts so that its volume V decreases at a rate proportional to its surface area S (and also that it stays spherical as it melts).
- (i) If r is its radius in centimetres at time t hours, show that the rate of change of r is constant. 1
- (ii) Given that it takes 3 hours for the snowball to decrease to half its original volume, show that $r = kt + R$, where R is the initial radius and $k = \frac{R}{3} \left(\frac{1}{\sqrt[3]{2}} - 1 \right)$. 3
- (iii) How much longer will it take for the snowball to melt completely? 1
- (d) Find the value of the following limit: $\lim_{x \rightarrow -5} \frac{\sqrt{20-x} - 5}{5+x}$ 2

Question 14 (15 marks) Start a NEW page

(a) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola $x^2 = 4ay$.

(i) Find the coordinates of the point of intersection T of the tangents to the parabola at P and Q .

2

(ii) Given that the tangents at P and Q intersect at 45° show that

$$p - q = 1 + pq \text{ where } p > q.$$

2

(iii) Find the equation of the locus of T when the tangents at P and Q intersect as given in (ii).

2

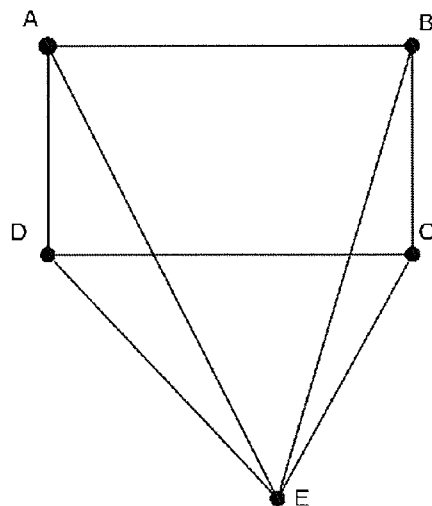
(b) A plane flying horizontally at a speed of 180 km/h is observed from point E on the ground (see diagram). Initially it is observed to be at A , on a bearing 337°T and at an elevation of 34° . After 2 minutes it is observed to be at B , bearing 018°T at an elevation of 27° .

(i) Show that $\angle DEC = 41^\circ$

1

(ii) Find the height h (in metres) of the plane.

3



(c) Given the function $f(x) = e^x + e^{2x}$,

(i) Write down the domain and range of $f(x)$.

1

(ii) Hence write down the domain and range of its inverse function $f^{-1}(x)$.

1

(iii) Find the equation of $f^{-1}(x)$.

3

END OF EXAMINATION

Multiple Choice

① B

⑥ B

② A

⑦ B

③ A

⑧ C

④ B

⑨ B

⑤ C

⑩ B

10

Q 11 (a) $\lim_{x \rightarrow 0} \frac{\sin 2x \cos 2x}{3x} = \frac{2}{3}$ ✓

(b) $\cos^{-1} \cos \left(\frac{5\pi}{4} \right) = \frac{3\pi}{4}$ ✓

(c) $\int \cos^2 9x \, dx = \int \frac{1}{2} (1 + \cos 18x) \, dx$
 $= \frac{x}{2} + \frac{\sin 18x}{36} + C$ ✓

(d) $\sin x + \cos x = \sqrt{2} \sin \left(x + \frac{\pi}{4} \right)$ ✓

$\therefore \sqrt{2} \sin \left(x + \frac{\pi}{4} \right) = 1, \quad 0 \leq x \leq 2\pi$

$\therefore \sin \left(x + \frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}, \quad \frac{\pi}{4} \leq x + \frac{\pi}{4} \leq 2\pi + \frac{\pi}{4}$

$\therefore x + \frac{\pi}{4} = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{\pi}{4} + 2\pi$

$\therefore x = 0, \frac{\pi}{2}, 2\pi.$ ✓✓

[alt: can use t-method:

$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} = 1 \rightarrow t = 0 \text{ or } 1$

$\rightarrow \tan \frac{x}{2} = 0 \text{ or } 1$

$\therefore \frac{x}{2} = 0, \pi, \frac{\pi}{4}$

$\therefore x = 0, 2\pi, \frac{\pi}{2}$

⊛ Must test $x = \pi$

← (*)

Q11 ctd.

(e) $P(x) = (1-x)(1+x) \cdot Q(x) + 4-x$

remainder when divided by $1+x$, i.e. by $x+1$,
is $P(-1)$

$$\therefore \text{rem.} = P(-1) = 4 - (-1) = 5$$

✓✓

(f) $P(x) = x^3 - 6x^2 + 3x + k = 0$

let roots be $a-d$, a , $a+d$

$$\therefore \text{sum roots} = 3a = 6 \quad \therefore a = 2$$

✓

$$\therefore \text{Since } 2 \text{ is a root, } P(2) = 0$$

$$\therefore 2^3 - 6 \cdot 2^2 + 3 \cdot 2 + k = 0$$

$$-10 + k = 0$$

$$\therefore k = 10$$

✓✓

(g) When $n=1$, $n^3 + (n+1)^3 + (n+2)^3 = 1^3 + 2^3 + 3^3$
 $= 36$
 $= 9 \times 4$
 $\therefore \text{true for } n=1.$

✓

Assume true for $n=k$,

i.e. assume $k^3 + (k+1)^3 + (k+2)^3 = 9M$, M an integer.

Then $(k+1)^3 + (k+2)^3 + (k+3)^3 = (9M - k^3) + (k+3)^3$,
by assumption

✓

$$= 9M - \cancel{k^3} + \cancel{k^3} + 3 \cdot k^2 \cdot 3 + 3 \cdot k \cdot 3 + 3^3$$

$$= 9M + 9k^2 + 27k + 27$$

$$= 9 [M + k^2 + 3k + 3]$$

$$= 9N, \quad N \text{ an integer,} \\ \text{as } M \text{ \& } k \text{ are integers.}$$

$$\therefore \text{true for } n=k+1 \text{ if true for } n=k.$$

✓ (no fudging!)

Since true for $n=1$, $\therefore$ true for $n=1, 2, 3, \dots$
by induction.

Q12

$$(a) \quad I = \int_3^8 \frac{x-1}{\sqrt{x+1}} dx$$

$$\text{Let } x = u^2 - 1, \quad u > 0.$$

$$\begin{cases} x=3 \rightarrow u=2 \\ x=8 \rightarrow u=3 \end{cases}$$

$$\therefore dx = 2u \, du$$

$$\begin{cases} x-1 = u^2 - 2 \\ \sqrt{x+1} = u \end{cases}$$

$$\therefore I = \int_2^3 \frac{u^2 - 2}{u} \cdot 2u \, du$$

$$= 2 \int_2^3 (u^2 - 2) \, du$$

$$= 2 \left[\frac{u^3}{3} - 2u \right]_2^3 = 2 \left[\left(\frac{27}{3} - 6 \right) - \left(\frac{8}{3} - 4 \right) \right]$$

$$= \frac{26}{3}$$

$$(b) \quad (i) \quad T = 22 - Ae^{-kt} \quad \therefore \frac{dT}{dt} = KAe^{-kt}$$

$$= K(22 - T)$$

$$= -k(T - 22)$$

$$\text{When } t=0, T=8 \quad \therefore -8 = 22 - A \quad \therefore A = 30$$

$$(ii) \quad \text{When } t=90, T=4 \quad \therefore 4 = 22 - 30e^{-90k}$$

$$\therefore e^{-90k} = \frac{18}{30} = \frac{3}{5}$$

$$\therefore -90k = \ln \frac{3}{5} \quad \therefore k = -\frac{1}{90} \ln \left(\frac{3}{5} \right)$$

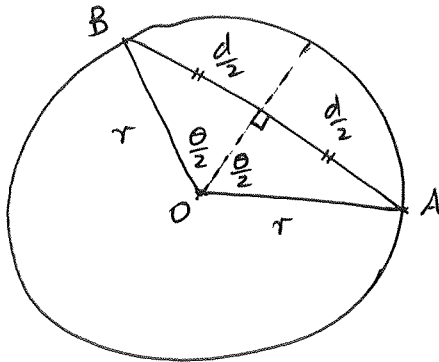
$$(iii) \quad \text{When } t=180, \quad -180k$$

$$T = 22 - 30e^{-180k}$$

$$= 22 - 30e^{2 \ln \left(\frac{3}{5} \right)} = 22 - 30 \times \frac{9}{25} = 11.2^\circ \text{C}$$

Q12-ctd

(c)(i)



Join O to the midpoint of AB.
It meets it at 90° (circle
geom. theorem)

$$\therefore \sin \frac{\theta}{2} = \frac{\frac{d}{2}}{r} = \frac{d}{2r}$$

$$\therefore d = 2r \sin\left(\frac{\theta}{2}\right)$$

Also, $L = r\theta$

Since $\frac{L}{d} = \frac{4}{3}$

$$\therefore \frac{\cancel{r}\theta}{2\cancel{r}\sin\left(\frac{\theta}{2}\right)} = \frac{4}{3}$$

$$\therefore 3\theta = 8 \sin\left(\frac{\theta}{2}\right)$$

(ii) Let $f(\theta) = 3\theta - 8 \sin \frac{\theta}{2}$

then $f'(\theta) = 3 - 4 \cos \frac{\theta}{2}$

Thus $\theta_2 = \theta_1 - \frac{f(\theta_1)}{f'(\theta_1)}$

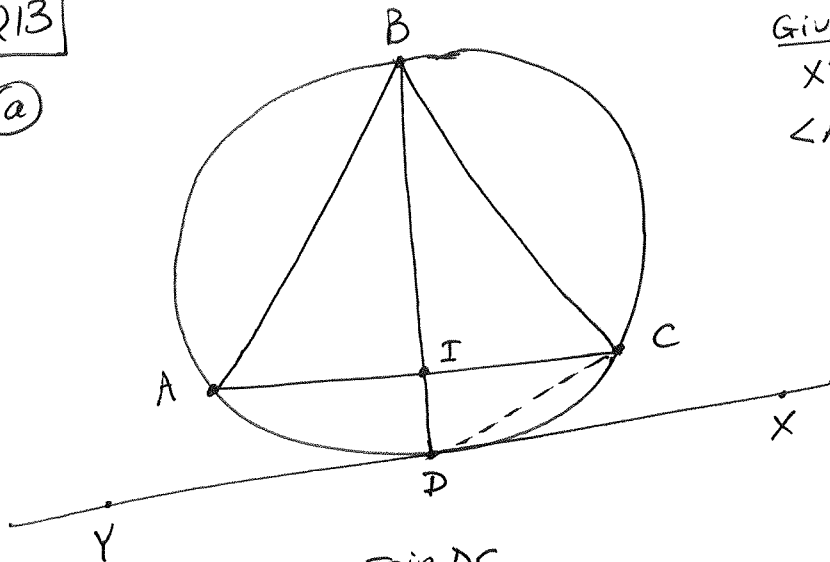
$$= 2.5 - \frac{3(2.5) - 8 \sin(1.25)}{3 - 4 \cos 1.25}$$

$$\approx 2.5528420 \dots$$

$\therefore \theta_2 = 2.553$ (3 d.p.)

Q13

(a)

Given:

XY is tangent,
 $\angle ABD + \angle BCA = 90^\circ$

Join DC.

$$(i) \quad \angle BCD = \angle BCA + \angle ACD.$$

But $\angle ACD = \angle ABD$ (angles in the same segment)

$$\therefore \angle BCD = \angle BCA + \angle ABD \\ = 90^\circ \text{ (given)}$$

Hence BD is a diameter (converse of "angle in a semicircle is 90° " theorem)

$$(ii) \quad \text{Since } AB = BC,$$

$$\therefore \angle BAC = \angle BCA \text{ (equal angles opposite to equal sides in } \triangle ABC)$$

$$\text{So } \angle AID = \angle BAC + \angle ABD \text{ (exterior angle of triangle equals sum of opposite interior angles)}$$

$$= \angle BCA + \angle ABD$$

$$= 90^\circ \text{ (from (i))}$$

$$\text{But } \angle IDX = 90^\circ \text{ (tangent is perpendicular to radius to point of contact)}$$

$$\therefore AC \parallel XY \text{ (alternate angles equal)}$$

[note: alternate routes possible to solution here].

Q13 ctd

$$(b) (i) \quad v = 1+x^2 \quad \therefore \quad \frac{1}{2}v^2 = \frac{1}{2}(1+x^2)^2$$

$$\therefore a = \frac{d}{dx} \left(\frac{1}{2}v^2 \right) = \frac{1}{2} \cdot 2 \cdot 2x(1+x^2)$$

$$\therefore a = 2x(1+x^2)$$

✓✓

$$(ii) \quad \frac{dx}{dt} = 1+x^2 \quad \therefore \quad \frac{dt}{dx} = \frac{1}{1+x^2}$$

$$\therefore t = \tan^{-1}x + C$$

✓

$$\text{at } t=0, x=1 \quad \therefore \quad 0 = \tan^{-1}1 + C$$

$$= \frac{\pi}{4} + C \quad \therefore \quad C = -\frac{\pi}{4}$$

$$\therefore t = \tan^{-1}x - \frac{\pi}{4}$$

$$\therefore \tan^{-1}x = t + \frac{\pi}{4}$$

$$\therefore x = \tan\left(t + \frac{\pi}{4}\right)$$

✓

(c)

$$\frac{dV}{dt} \propto S \quad \therefore \quad \frac{dV}{dt} = kS, \text{ for some constant } k.$$

$$\text{But } \frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$\therefore kS = 4\pi r^2 \cdot \frac{dr}{dt} \quad \left(\text{as } V = \frac{4}{3}\pi r^3\right)$$

$$\text{But } S = 4\pi r^2$$

$$\therefore k = \frac{dr}{dt}$$

$$\text{ie } \frac{dr}{dt} \text{ is constant.}$$

✓

Q13 ctd

(c) (ii) from (i), $\frac{dr}{dt} = k$

$$\therefore r = kt + c$$

$$\text{at } t=0, r=R \quad \therefore R = 0 + c \quad \therefore c = R$$

$$\therefore r = kt + R$$

$$\therefore \text{at } t=3, r = 3k + R$$

$$\text{So we have: } \frac{4}{3}\pi(3k+R)^3 = \frac{1}{2} \cdot \frac{4}{3}\pi R^3 \quad (\text{from the given data})$$

$$3k+R = \frac{1}{\sqrt[3]{2}} \cdot R$$

$$3k = R\left(\frac{1}{\sqrt[3]{2}} - 1\right) \quad \therefore k = \frac{R}{3}\left(\frac{1}{\sqrt[3]{2}} - 1\right)$$

$$(iii) \quad r = \frac{R}{3}\left(\frac{1}{\sqrt[3]{2}} - 1\right)t + R$$

$$\text{When fully melted, } r=0$$

$$\therefore \frac{1}{3}\left[\frac{1}{\sqrt[3]{2}} - 1\right]t + 1 = 0$$

$$\therefore t\left(\frac{1}{\sqrt[3]{2}} - 1\right) = -3 \quad \therefore t = \frac{-3}{\left(\frac{1}{\sqrt[3]{2}} - 1\right)} \doteq 14.54196\dots$$

$$\text{So extra time taken is } \doteq 11.54 \text{ hours} \\ (\text{i.e. } 11 \text{ h } 33 \text{ mins}).$$

$$(d) \quad \lim_{x \rightarrow -5} \frac{\sqrt{20-x} - 5}{5+x} = \lim_{x \rightarrow -5} \frac{\sqrt{20-x} - 5}{5+x} \times \frac{\sqrt{20-x} + 5}{\sqrt{20-x} + 5}$$

$$= \lim_{x \rightarrow -5} \frac{20-x-25}{(5+x)\sqrt{20-x}+5}$$

$$= \lim_{x \rightarrow -5} \frac{-1}{\sqrt{20-x}+5} = -\frac{1}{10}$$

(needs working
for full
marks)

Q14 (a)

(i) Using the formula sheet, at T we have

$$px - ap^2 = qx - aq^2$$

$$(p-q)x = ap^2 - aq^2$$

$$\therefore x = a(p+q) \quad \text{as } p \neq q$$

$$\therefore y = pa(p+q) - ap^2 = apq$$

$$\therefore T = [a(p+q), apq]$$

(some working required)

✓✓

(ii) The tangents have gradients p, q .

$$\therefore \tan 45^\circ = \left| \frac{p-q}{1+pq} \right|$$

$$\therefore 1 = \frac{p-q}{1+pq} \quad (p > q)$$

$$\text{ie. } p-q = 1+pq$$

✓ needs:
 • $\tan 45^\circ$
 • abs. value.
 • $p > q$

✓

$$\text{(iii) from (i), } x = a(p+q) \quad \therefore p+q = \frac{x}{a}$$

$$y = apq \quad \therefore pq = \frac{y}{a}$$

$$\text{Now } (p-q)^2 = (p+q)^2 - 4pq$$

$$\therefore (1+pq)^2 = (p+q)^2 - 4pq \quad \text{from (ii)}$$

$$\therefore \left(1 + \frac{y}{a}\right)^2 = \left(\frac{x}{a}\right)^2 - \frac{4y}{a}$$

$$1 + \frac{2y}{a} + \frac{y^2}{a^2} = \frac{x^2}{a^2} - \frac{4y}{a}$$

$$\frac{x^2}{a^2} = \frac{y^2}{a^2} + 1 + \frac{6y}{a}$$

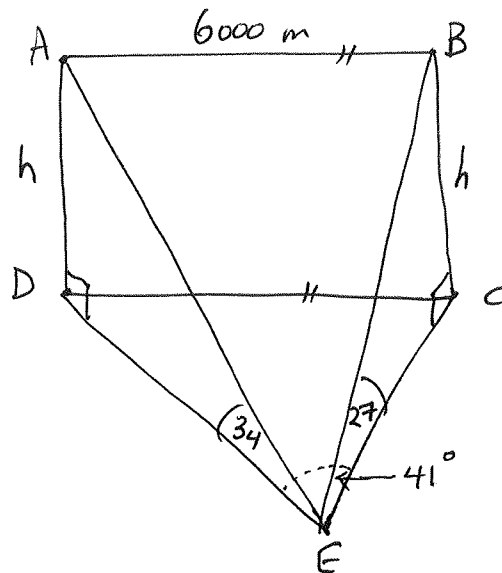
$$\text{ie } x^2 = y^2 + a^2 + 6ay$$

✓ uses
 part (ii)
 in way that
 could lead
 to solution

✓ or
 equivalent
 expression

Q14 - ctd.

(b)



- (i) The bearing of A (& hence of D) is 337°T ,
& the bearing of B (& hence of C) is 018°T .
 $\therefore \angle DEC = (360^\circ - 337^\circ) + 018^\circ$
 $= 23^\circ + 18^\circ = 41^\circ$

(must not
assume
flies W-E)

✓

(ii) $\frac{h}{DE} = \tan 34^\circ \therefore DE = \frac{h}{\tan 34^\circ} \therefore DE = h \cot 34^\circ$
 Likewise, $CE = h \cot 27^\circ$

✓

(in 2 mins @ 180 km/h, it goes 6 km = 6000 m)

By cosine rule in $\triangle DEC$,

$$6000^2 = h^2 \cot^2 34^\circ + h^2 \cot^2 27^\circ - 2h \cot 34^\circ \cdot h \cot 27^\circ \cdot \cos 41^\circ$$

$$= h^2 [\cot^2 34^\circ + \cot^2 27^\circ - 2 \cot 34^\circ \cot 27^\circ \cos 41^\circ]$$

(incorrect
Cosine rule
terminates)(cannot use
sine
rule)

✓

$$\therefore h = \frac{6000}{\sqrt{\cot^2 34^\circ + \cot^2 27^\circ - 2 \cot 34^\circ \cot 27^\circ \cos 41^\circ}}$$

$$= \frac{6000}{\sqrt{\tan^2 56^\circ + \tan^2 63^\circ - 2 \tan 56^\circ \tan 63^\circ \cos 41^\circ}}$$

$$\therefore h \doteq 4659.87 \text{ m}$$

$$h = 4660 \text{ m (nearest metre)}$$

✓

Q14-ctd.

$$(c) \quad f(x) = e^x + e^{2x}, \quad \text{i.e. } y = e^x + e^{2x}$$

$$(i) \quad \text{For } f, \quad D: \text{all real } x \\ R: y > 0$$

$$(ii) \quad \text{For } f^{-1}, \quad D: x > 0 \\ R: \text{all real } y$$

$$(iii) \quad \text{for inverse, } x = e^y + e^{2y}$$

$$\text{i.e. } e^{2y} + e^y - x = 0$$

$$\text{let } u = e^y$$

$$\therefore u^2 + u - x = 0$$

$$u = \frac{-1 \pm \sqrt{1+4x}}{2}$$

$$\therefore e^y = \frac{-1 + \sqrt{1+4x}}{2} \quad \text{or} \quad \frac{-1 - \sqrt{1+4x}}{2}$$

$$\text{But } e^y > 0 \quad \therefore \quad e^y = \frac{-1 + \sqrt{1+4x}}{2}$$

$$\therefore y = \ln \left[\frac{\sqrt{1+4x} - 1}{2} \right]$$

must reject
one solution